Chapter-2

Microprocessor Architecture and Microcomputer System

Every modern Intel (like the Core i9) and AMD Ryzen chip is a microprocessor (MPU).

Microprocessor

❑ A microprocessor is programmable logic device, designed with registers, flip-flops, and timing elements. It is a single chip made by LSI or VLSI technology. It is the CPU of a microcomputer.

Microprocessor Unit (MPU)

❑In early processors the different units of a CPU are made not in a single chip rather in different separate chips. These group of chips are called a Micro Processor Unit (MPU).

e.g., separate ALU chip, separate control unit chip, separate register chips

2.1 MICROPROCESSOR ARCHITECTURE AND ITS OPERATIONS

Architecture of Microprocessor

- The internal logic diagram of a microprocessor is called its architecture.

Different Categories of Functions Performed by a Microprocessor

All the various functions performed by the microprocessor can be classified in three categories:

- Microprocessor-initiated operations.
- Internal data operations.
- Peripheral (or externally) initiated operations.

The CPU is the boss. It actively goes out to read from or write to memory and other devices.

2.1.1 Microprocessor-Initiated Operations

The MPU performs primarily four operations:

- Memory Read: Reads data from memory.
- Memory Write: Writes data into memory.
- I/O Read: Accepts data from input devices.
- I/O Write: Sends data to output devices.

Steps to Communicate with Peripherals or Memory Location

To communicate with a peripheral or a memory location, the MPU needs to perform the following steps:

- Step 1: Identify the peripheral or the memory location (with its address)
- Step 2: Transfer data.
- Step 3: Provide timing or synchronization signals.

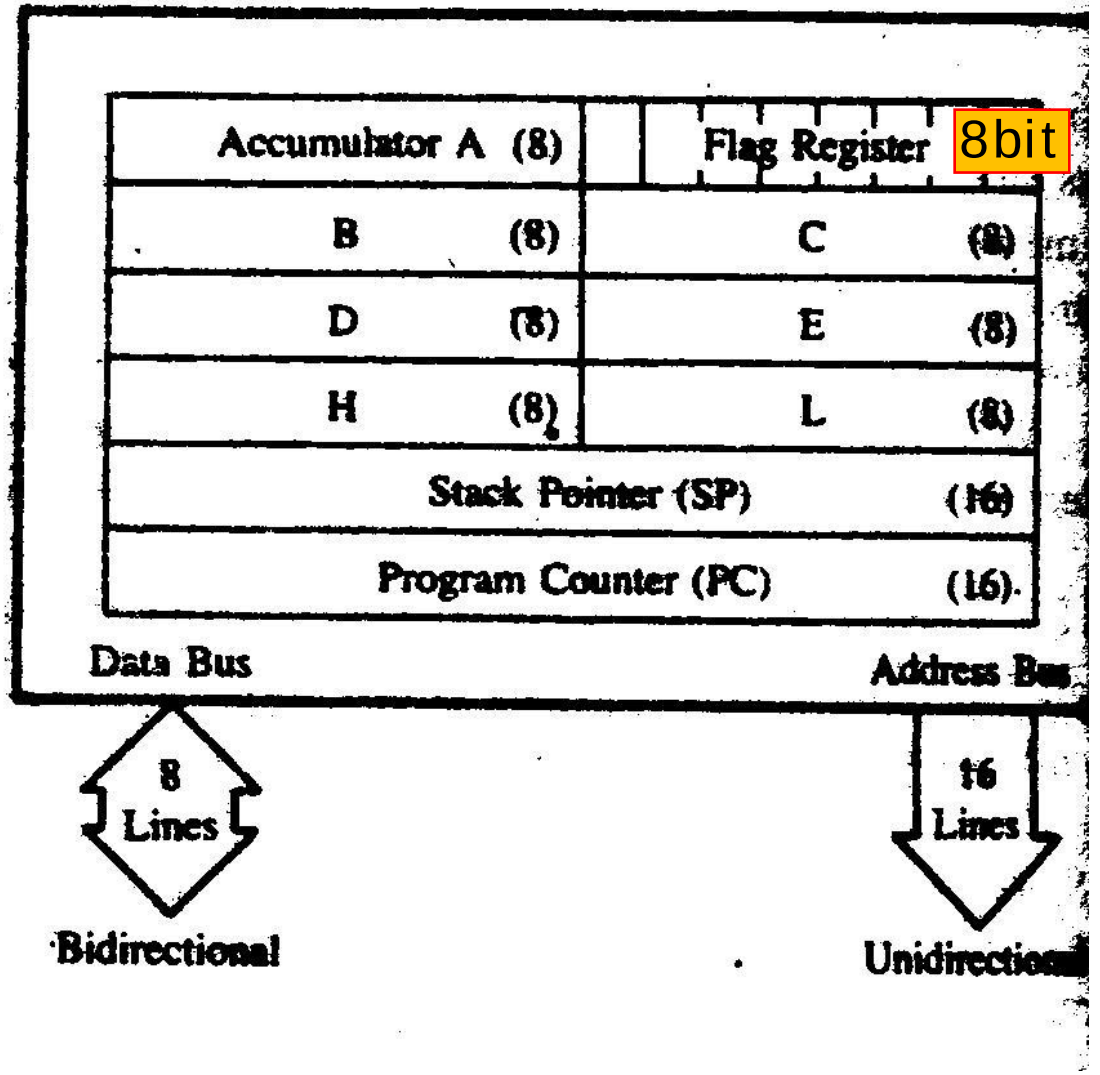
2.1.2 Internal Data Operations and the 8085/8080A Registers

The internal architecture of the 8085/8080A microprocessor determines how and what operations can be performed with data. These operations are: allow to perform five main types of operations

- Store 8-bit data
- Perform arithmetic and logical operations.
- Test for conditions. can check any flags conditons
- Sequence the execution of instructions. must execute instructions in correct order
- Store data temporarily during execution in the defined R/W memory locations called the stack.

Internal Data Operation explain what the microprocessor can do with data internally

FIGURE 2.3
The 8085 Programmable Registers



2.1.3 Peripheral or Externally Initiated Operations

External devices (or signals) can initiate the following operations for which individual pins on the microprocessor chip are assigned:

- **Reset:** When all internal operations are suspended and the program counter(PC) is cleared (0000H). Full Processor Restart
- **Interrupt:** Microprocessor can be interrupted from the normal execution and asked to execute some other instructions called service routine. After completing service routine it resumes its operation. Just Pause the processor
- **Ready:** If the signal of 8085/8080A READY pin is low, the microprocessor enters into a Wait state. The pin is used to synchronize with slower peripherals with microprocessor.
- **Hold:** When the HOLD is activated by an external signal, the microprocessor relinquishes control of system bus for external peripheral to use them. For example, HOLD signal is used in DMA data transfer.

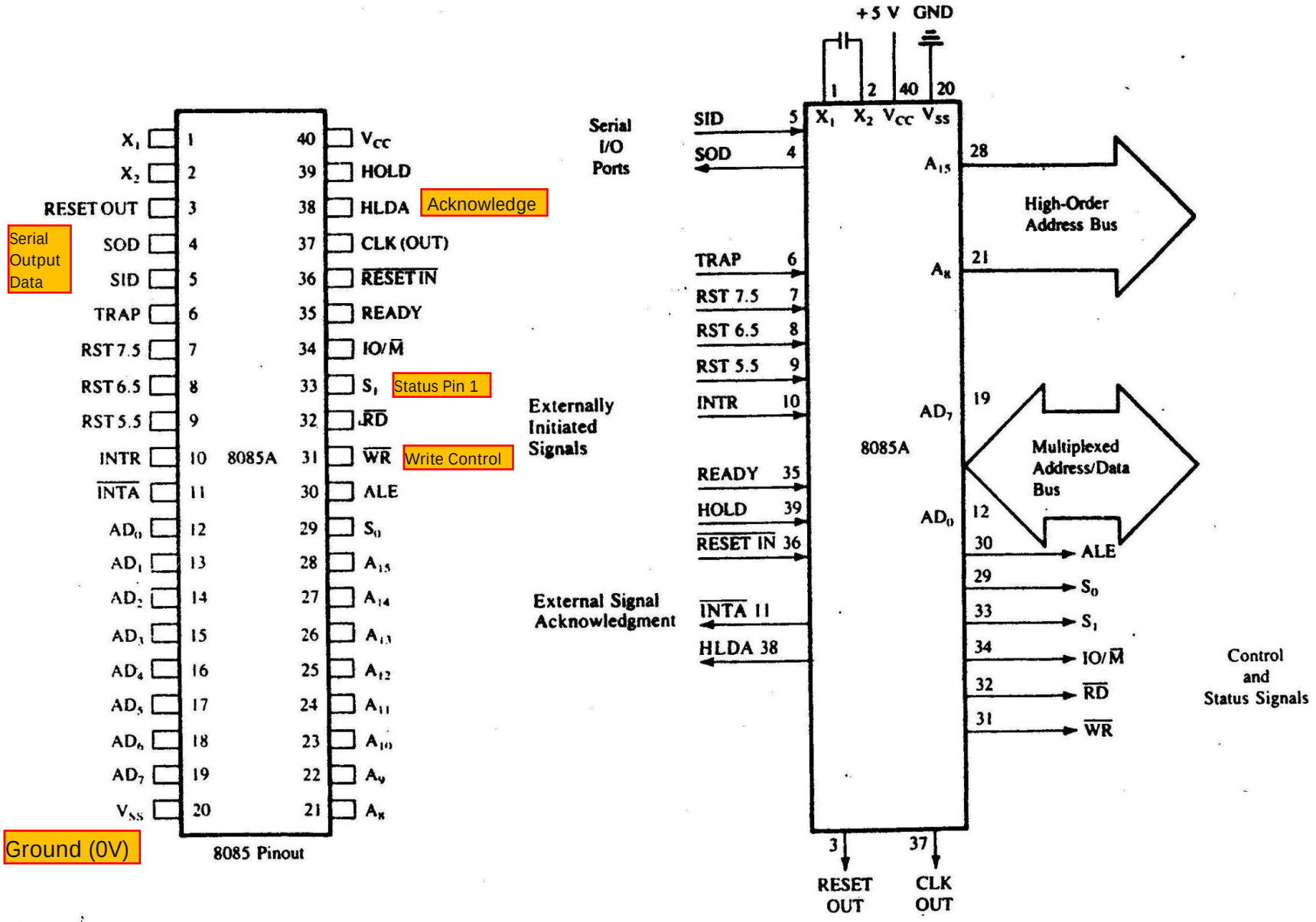


FIGURE 3.1
The 8085 Microprocessor Pinout and Signals

NOTE: The 8085A is commonly known as the 8085.
SOURCE (Pinout): Intel Corporation, *Embedded Microprocessors* (Santa Clara, Calif.: Author, 1994), pp. 1-11.